An Elementary Exploration of Kalman Filtering for Tracking: Studies In Corner-Case Dynamics

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ASPIRE

Background

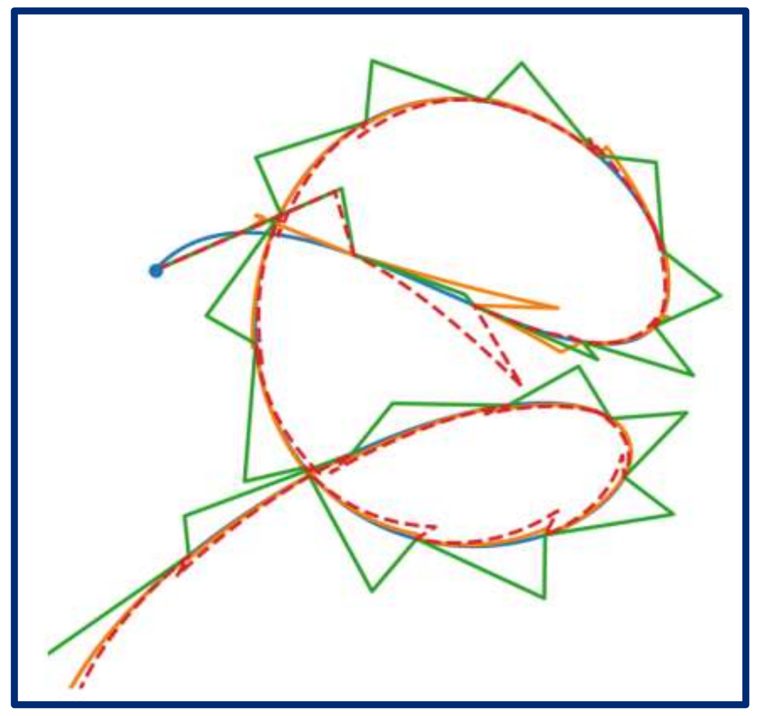
Kalman Filters are a widely implemented and powerful family of state estimators, capable of **updating an internal state estimate** with, often noisy, **measurement data**. In addition to predicting and correcting the estimates of the state vector, the filter upkeeps a **covariance matrix** which defines the magnitude and relationship of **uncertainties within the state parameters**.

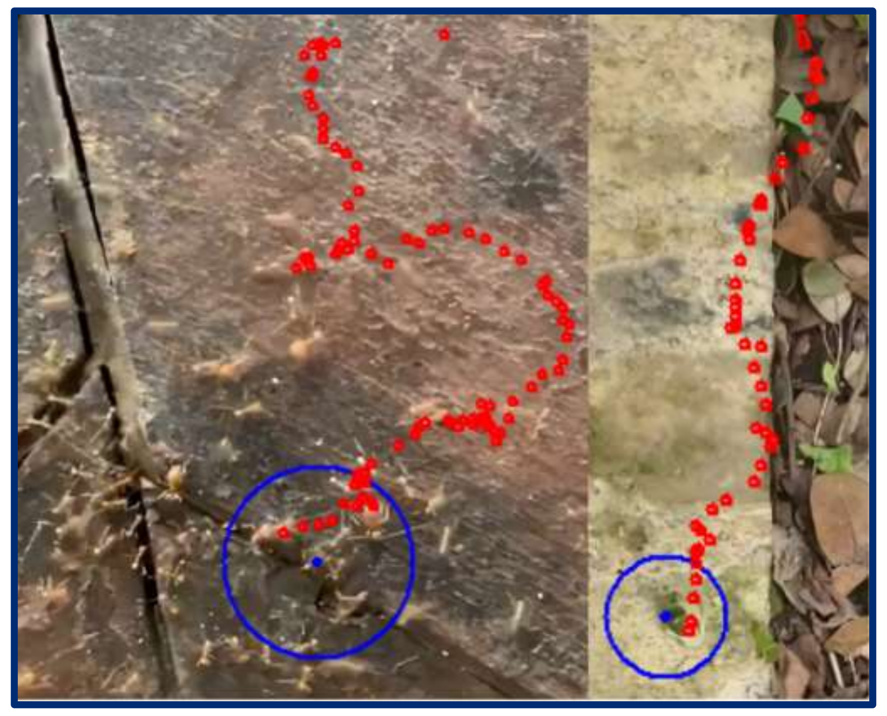
In the context of tracking, the Kalman Filter is used to **establish and update the true location, motion, and probable region** of a maneuvering target as accurately as possible, **even in the presence of noise**. Updates may come from a Radar Sensor or the like, which may only provide partial state information.

Outcomes

- Coded Python **implementations** of the Kalman Filter and the Square Root Information Filter.
- Developed custom Python classes to **simulate, update, visualize, and quantify performance** of various Kalman Filter tracking models and methods.
- Studied the **effects of manipulating model dynamics**, such as those associated with circular trajectories or possessing high-order kinematic equations, on **filter accuracy and effectiveness**.
- Examined practical methods for **optimizing filter performance,** specifically in the context of **covariance growth** and **data rate**.
- Learned a great deal about the **problems and techniques** pertinent to the field of **tracking and Kalman Filtering**.

Simulation Demonstrations





*Left: A noise-less simulation of a **winding trajectory tracked by low-data-rate linear and helical Kalman Filter models**. Each 'spike' corresponds to an **update** to the **maximum likelihood estimate** of the target particle.*

*Right: Position-Velocity Kalman tracking of ant behavior alongside a **visualization of the covariance bubble;** the first a chaotic colony and the second a stable, linear march. These differences in particle motion were key to studying **the differences in performance of Position-Only and Position-Velocity trackers**.*

Relation to Past and Future Work

The Kalman Filter plays a central role in **Multiple Hypothesis Tracking** and probabilistic **track assignment** as the means of computing the likelihood of a particular association of measurements or a hypothesis of target motion. The (Log-Likelihood) entries to these algorithms are given directly from the **residuals and covariance of the Kalman filter.**

The previous ASPIRE session was dedicated to studying the Permanent and Expected Assignment of **cost matrices,** whose entries are obtained via Kalman Filtering techniques. We continue to examine the complexities of the track association problem in the context of **Multidimensional Assignment**, and a better understanding of the **Kalman Filter architecture and the covariance it produces** will aid in this exploration.

Filtering Overview & Research Questions

How can we alter the **underlying dynamical system** of the filter to best suit application?

Internal Motion Model

- Studied the behavior and construction of higher-order kinematic motion models and evaluated their performance.
- Developed a framework of Helical Models, combining the ability to track both linear and circular trajectories.

$$\Phi_{ij}^{(k)}(\Delta t) = \begin{cases} \dfrac{(\Delta t)^{(k-i)-(k-j)}}{[(k-i)-(k-j)]!}, & 0 \le i \le j \le k-1 \\ 0, & Otherwise \end{cases}$$

$$Q_{ij}(\Delta t) = \left[\int_0^{\Delta t} (\Phi(t)\vec{q})(\Phi(t)\vec{q})^T\, dt\right]_{ij} = \frac{1}{[k-i-1]!\,[k-j-1]!}\,\frac{(\Delta t)^{(k-i)+(k-j)-1}}{(k-i)+(k-j)-1}$$

Example: Formulas for the propagation and process noise matrices, Phi and Q, for a kinematic model with k derivative parameters. The Helical Model bears mixed sinusoid and polynomial entries, for instance.

Kalman Filter

Process Noise

Partial State Updates

Data Rate

How can we **optimize** the update scheme and **manage the growth of uncertainty** in the presence of **noise**?

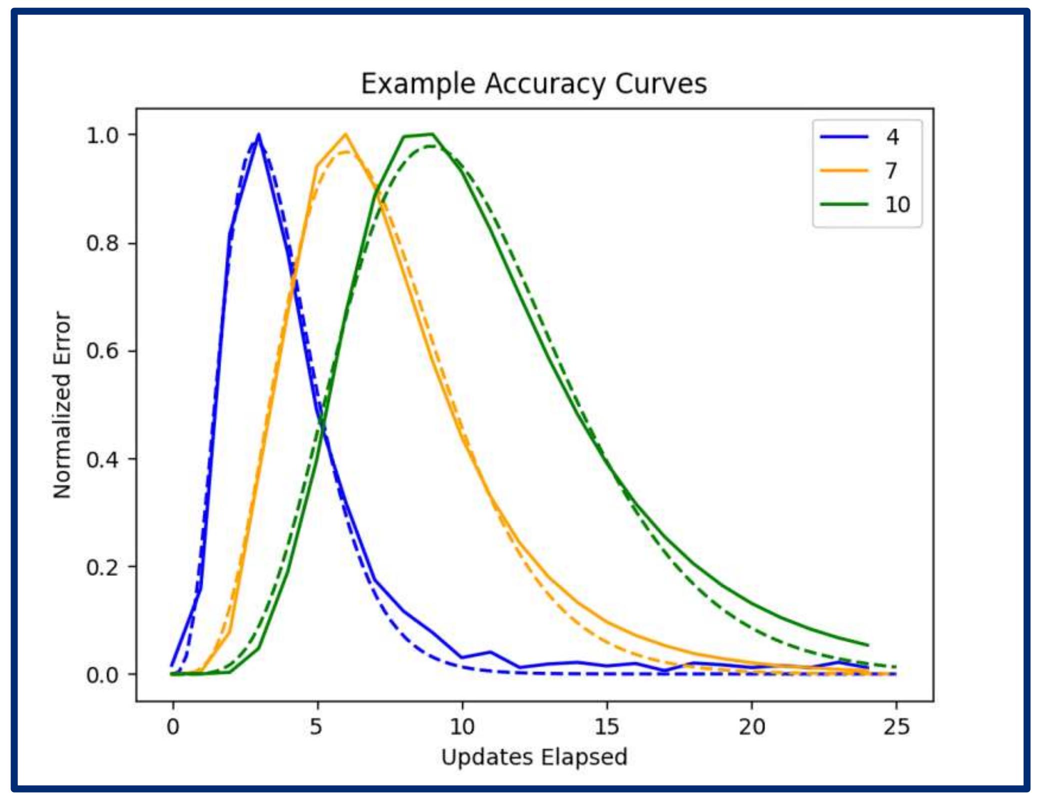
- Examined the effect of a variable update rate as well as variably optimized process noise on tracking effectiveness.

How can we **regularize** and **transform** the covariance bubble to reflect real-world uncertainty?

Covariance Updates

Full State Estimation

- Implemented eigenvalue regularization and preferential shearing of the covariance ellipse.
- Investigated the advantages and tradeoffs of increasing state parameters on model convergence and tracking accuracy.



Convergence of error as a function of time for models estimating increasingly more state derivatives (Degree 4,7,10).

Tunable Parameters

Filtering of Measurement Inputs

Target Hypotheses

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